

DSA Problem Recognition Cheat Sheet

Master pattern recognition to map clues to optimal approaches instantly

The 22 Essential DSA Patterns

1. Search in Unsorted Array/List

Clues: Array is not sorted. Find/check/count an element.

Think: **Linear Search, HashMap / HashSet**

Examples: Two Sum, Contains Duplicate

2. Search in Sorted Array/List

Clues: Array is sorted. Need fast search.

Think: **Binary Search**

Examples: Binary Search, Search Insert Position

3. Pair Problems

Clues: Find two numbers. Pair sum/difference.

Think: **Hashing (unsorted), Two Pointers (sorted)**

Examples: Two Sum, 167. Two Sum II

4. Triplet / Quadruplet Problems

Clues: Find 3 or 4 elements satisfying a condition.

Think: **Sort + Two Pointers**

Examples: 3Sum, 4Sum

5. Contiguous Subarray / Substring

Clues: Keywords: 'Subarray', 'Substring', 'Longest', 'Shortest'.

Think: **Sliding Window, Prefix Sum**

Examples: Max Average Subarray, Min Window Substring

6. Range Sum Queries

Clues: Multiple range queries. Sum from L to R.

Think: **Prefix Sum**

Examples: Range Sum Query

7. Nearest Greater/Smaller Element

Clues: Keywords: Next Greater, Previous Smaller, Nearest Bigger.

Think: **Monotonic Stack**

Examples: Daily Temperatures, Next Greater Element

8. Frequent Min/Max Extraction

Clues: Repeatedly find largest/smallest. Top K.

Think: **Heap / Priority Queue**

Examples: Kth Largest Element, Top K Frequent Elements

9. All Possible Combinations

Clues: Keywords: Generate all, All subsets, All permutations.

Think: **Backtracking**

Examples: Subsets, Permutations

10. Decision Problems

Clues: Minimize/Maximize something. 'What is the minimum X?'

Think: **Binary Search on Answer**

Examples: Koko Eating Bananas, Ship Packages Within D Days

11. Dependency Problems

Clues: Keywords: Prerequisite, Dependency, Course order.

Think: **Topological Sort**

Examples: Course Schedule

12. Connectivity Problems

Clues: Keywords: Connected components, Number of groups/provinces.

Think: **DFS/BFS, DSU**

Examples: Number of Provinces

13. Shortest Path Problems

Clues: Keywords: Minimum steps, Shortest path.

Think: **BFS (unweighted), Dijkstra (weighted)**

Examples: Word Ladder, Network Delay Time

14. Matrix Traversal Problems

Clues: Grid/2D matrix. Explore neighbours.

Think: **DFS/BFS**

Examples: Number of Islands, Flood Fill

15. Tree Problems

Clues: Keywords: Binary Tree, Path, Ancestor.

Think: **DFS, Recursion**

Examples: Maximum Depth, Path Sum

16. Repeated Subproblems

Clues: Recursion causes recomputation. Optimize recursive solution.

Think: **Dynamic Programming**

Examples: Climbing Stairs, Coin Change

17. String Matching

Clues: Keywords: Pattern matching, Prefix/Suffix.

Think: **KMP, Sliding Window, Trie**

Examples: Implement strStr()

18. Interval Problems

Clues: Keywords: Meetings, Ranges, Overlapping intervals.

Think: **Sort + Merge**

Examples: Merge Intervals, Meeting Rooms

19. Cycle Detection / Middle Element

Clues: Linked List cycle. Find middle. Duplicate number ($O(1)$ space).

Think: **Fast and Slow Pointers**

Examples: Linked List Cycle, Middle of Linked List

20. Word Dictionary / Prefix Matching

Clues: Search words quickly. Autocomplete. Prefix matching.

Think: **Trie (Prefix Tree)**

Examples: Implement Trie, Word Search II

21. Constant Space / Pairing Logic

Clues: Array has pairs, one is single. $O(1)$ space, $O(N)$ time.

Think: **Bit Manipulation (XOR)**

Examples: Single Number, Missing Number

22. Local Optimum Choice

Clues: Jump games, scheduling, maximize profit at current step.

Think: **Greedy Algorithm**

Examples: Jump Game, Buy and Sell Stock

Golden Interview Questions

Whenever you read a problem, ask yourself these 10 fundamental questions to immediately identify the core approach:

1. Is the data sorted?	Binary Search / Two Pointers
2. Need fast lookup?	HashMap / HashSet
3. Subarray/Substring?	Sliding Window / Prefix Sum
4. Need all possibilities?	Backtracking
5. Need optimal answer?	Greedy / DP / Binary Search on Answer
6. Need repeated min/max?	Heap
7. Graph/Tree/Grid?	DFS / BFS
8. Overlapping ranges?	Sort + Merge Intervals
9. Next greater/smaller?	Monotonic Stack
10. Dependencies?	Topological Sort

Pro Tip: Master these recognition tricks, and you will be able to identify the approach for most interview problems within 1–2 minutes.